

CONFIDENTIAL: BANK IT ARCHITECTURE & BUSINESS OPERATIONS BLUEPRINT

Document Type: Technical & Business Operations Blueprint

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Target Audience: AI Gap Analysis / System Architecture Review

Date: June 2026

1. Executive Summary

This document outlines the end-to-end IT architecture and business revenue models of a modern banking institution. It details how the bank generates revenue across retail and corporate sectors, the underlying technical infrastructure that supports high-frequency financial transactions, data storage mechanisms, and internal IT operations.

2. Business Model: Revenue Generation

The financial institution operates on a dual-pronged approach, serving both everyday consumers and large-scale enterprises.

A. Retail Banking (Consumer Segment)

- **Net Interest Margin (NIM):** The core profit engine. The bank acquires funds via Current/Savings Accounts (CASA) and Fixed Deposits at a cost of 3-7%. These funds are deployed as Personal, Auto, and Home Loans at 10-18%.
- **Retail Fee Income:** Revenue generated through credit card annual fees, ATM inter-bank transaction charges, processing fees for loans, and penalties for late payments.
- **Wealth Management & Distribution:** Commissions earned by selling third-party mutual funds, life insurance, and gold products to existing retail clients.

B. Corporate Banking (B2B Segment)

- **Working Capital & Term Loans:** Large-scale credit facilities provided to businesses for expansion, raw material procurement, and operations.
- **Transaction Banking & Payment Gateways:** The bank processes millions of merchant transactions via PoS terminals and online gateways, capturing a 1-2% Merchant Discount

Rate (MDR) per transaction.

- **Forex & Trade Finance:** High-margin revenue generated from currency conversion, issuing Letters of Credit (LCs), and Bank Guarantees for international trade.
- **Banking-as-a-Service (BaaS):** Monetizing proprietary APIs by allowing corporate clients to integrate banking functions (like automated vendor payouts and payroll) directly into their ERP systems.

3. Technical Architecture

To ensure zero downtime, ACID compliance, and scalability, the IT infrastructure is divided into decoupled microservices.

A. Client-Facing Channels (Frontend)

- **Retail Platforms:** Native mobile applications and responsive web portals for individual users.
- **Corporate Platforms:** High-security web portals with multi-level maker-checker authorization workflows.

B. Middleware & API Gateway

- **Gateway Routing:** All incoming traffic routes through a secure API gateway managing SSL termination, rate-limiting, and DDOS protection.
- **Microservices (FastAPI/Spring Boot):** Business logic is divided into isolated services (e.g., Auth Service, Transaction Service, Notification Service). High-performance asynchronous frameworks like FastAPI are utilized to handle concurrent user requests efficiently.

C. Core Banking System (CBS)

- The system of record. Modernized CBS modules manage the global ledger, calculating interest, maintaining account balances, and ensuring transaction finality. It acts as the central hub connecting all peripheral applications.

4. Data Storage & Management

Financial data is highly sensitive and requires a tiered storage approach balancing speed, cost, and compliance.

Data Classification	Storage Technology	Primary Purpose
Hot Data (Transactional)	PostgreSQL / Oracle DB	Maintains the core ledger. Strict ACID compliance ensures money is never duplicated or lost during transfers.
Session & Cache	Redis	Stores short-lived data like login sessions, OTPs, and frequently accessed static configurations for ultra-fast retrieval.
Cold Data (Analytical)	AWS S3 Data Lakes	Archives historical transaction data, logs, and compliance records. Used heavily for regulatory reporting and ML model training.

5. Cloud Infrastructure & DevOps

The bank utilizes a hybrid infrastructure model, keeping highly sensitive core data on-premise while leveraging the cloud for scale.

- **Cloud Operations:** AWS is utilized for hosting non-core microservices, staging environments, and client-facing web applications.
- **CI/CD Pipeline:** The deployment lifecycle is fully automated. Code commits trigger Jenkins pipelines for automated testing, static code analysis, and vulnerability scanning.
- **Containerization:** Applications are packaged using Docker and orchestrated via Kubernetes (EKS), allowing the bank to auto-scale resources during peak transaction hours.

6. AI, Machine Learning & Risk Intelligence

Advanced data pipelines feed real-time transaction data into predictive models to safeguard the bank and optimize operations.

- **Real-time Fraud Detection:** Utilizing predictive modeling to analyze transaction velocity, location, and user behavior in milliseconds to block fraudulent transfers.

- **Credit Risk Scoring:** Machine learning algorithms assess non-traditional data points to generate accurate credit scores for users with thin financial histories.
- **Customer Support Automation:** NLP-driven virtual assistants handle Tier-1 customer queries (balance checks, card blocking), reducing operational overhead.

7. IT Operations & Hiring Structure

The IT department functions as a specialized technology firm bound by strict regulatory constraints.

1. **Software Development Engineers (SDEs):** Responsible for building scalable APIs, optimizing algorithms, and maintaining microservices.
2. **DevOps & Cloud Engineers:** Tasked with maintaining Jenkins pipelines, managing AWS infrastructure, and ensuring zero-downtime deployments via Docker/Kubernetes.
3. **Data Scientists & ML Engineers:** Focus on building predictive models, optimizing data lakes, and deploying intelligent decision engines.
4. **Information Security (InfoSec):** The most critical function. Security analysts perform continuous penetration testing, monitor networks for intrusions, and ensure compliance with global data protection regulations.